A two-stage Automated Claim-Checking System with an Evidence Fusion Verifier (EFV)

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Abstract

This report presents a two-stage automated fact-checking system designed to assess the veracity of claims by retrieving and evaluating supporting evidence from a given set of data. The system operates in two stages: (1) evidence retrieval, and (2) claim classification. For retrieval, we first implement a lexical baseline using BM25 to identify top-ranked evidence passages. To address the limitations of lexical overlap, we introduce dense retrieval approaches leveraging Sentence Transformers and FAISS, enabling efficient semantic search in high-dimensional vector space. We then leverage RoBERTa, a robustly optimised transformer model for claim verification as the next step. Finally, we implement an Evidence Fusion Verifier (EFV) to efficiently classify the validity of claims based on retrieved evidence.

1 Introduction

As we continue to live in a technological era where rapid and seamless communication of information is the new norm, the ability to discern factual claims from misinformation has become a critical skill. Automated fact-checking systems are becoming more prevalent in today's society, allowing users to distinguish claims based on its supporting evidences. In this research paper, we create a two-stage automated fact-checking system designed to assess the veracity of claims by (1) retrieving relevant evidence passages from a "knowledge source" and (2) classifying the claim based on the retrieved evidence passages.

Traditional approaches to fact-checking claims by manually reviewing evidence have been time-consuming and costly, often requiring domain-specific professionals to act as oracles, which could also introduce subjectivity into the process. As a consequence, manual verification is difficult to scale and is impractical for systems that must process large volumes of claims and evidence (Thorne et al., 2018). The automated system we propose addresses the challenge of accurately identifying supporting or refuting evidence while simultaneously handling cases where information may be insufficient or disputed.

We leveraged state-of-the-art Natural Language Processing (NLP) models and tools to effectively analyse potential similarities between a given claim and its potential evidence. We built a two-step system, composing of a hybrid retrieval system combining BM25 (lexical retriever) and semantic similarity retrievers, sentence transformers with FAISS and RoBERTa (a Meta AI powered transformer encoder). Our efforts extend to an Evidence Fusion Verifier (EFV) which builds on a standard DistilRoBERTa sentence-pair classifier evaluating multiple retrieved passages in a single pass.

Our work contributes to the growing field of research in automated fact-checking systems through the demonstration of different retrieval and classification techniques. We evaluated our system on a labeled dataset to ensure reproducibility and support effective benchmarking against standard approaches. Ultimately, our research further advances the development of reliable tools for detecting misinformation and other unfounded claims.

2 Approach

2.1 Preprocessing

A preprocessing pipeline was designed to standardise and refine the raw sentences into computable tokens for analysis. The approach consisted of the following steps:

- 1. Tokenisation and Conversion to Lowercase
- 2. Part of Speech (POS) Tagging
- 3. Lemmatisation
- 4. Stopword Removal

These preprocessing steps ensured that the claim and evidence sentences were accurate, efficient and consistent. Stopword removal removed any unnecessary noise to elevate the efficiency of the BM25 algorithm to focus on key terms. Lemmatisation and lowercase conversion ensured consistency in standardised form without losing important meaning.

2.2 Best Match 25 (BM25)

We adopted BM25 as an initial and lightweight lexical retriever for evidence passages, as it enables efficient ranking of the top k candidate evidence for a given claim. However, BM25's bag-of-words assumption (term independence) and lack of semantic modeling mean it can miss paraphrased or inferential links between claims and evidence. For a fact-checking task, claims and evidence often hinges on nuanced entailment (e.g. "X causes Y" vs. "Y causes X"). Pure lexical overlap may miss relevant evidence that uses different word choices, or surface-match irrelevant passages that happen to share the same words. As a result, BM25 would most likely be insufficient as our primary retriever.

2.3 Facebook AI Similarity Search (FAISS)

To address the limitations of BM25, we explored FAISS as another option as a retriever. By firstly encoding every evidence passage into a high-dimensional vector, using a pretrained Sentence-Transformer model ('all-MiniLM-L6-v2' as a starting point, and then normalizing those vectors. FAISS can then treat cosine similarity as an inner-product lookup. After the index was built, each claim was also embedded to be compared against the entire corpus (i.e., entire list of available passages), returning the top k passages whose vector representations lie closest to the claim (Douze et al., 2024). This dense approach excels at capturing paraphrases and conceptual relationships that BM25's term-independent scoring inevitably misses. For our relatively smaller datasets, we utilized FAISS' support for exact index (IndexFlatIP). In production, it was designed to mainly support approximate indices (e.g., IVF, HNSW, PQ) when dealing with millions of passages, which is ultimately where an automated fact-checking system should be deployed.

2.4 Hybrid Retriever

One unique approach that we implement when retrieving the top evidences for each claim for the list of evidences was the use of a hybrid retriever combining BM25 and sentence transformer. We initially use the BM25 to efficiently retrieve the top 30 candidate evidences for each given claim. However, as discussed above, while the BM25 algorithm excels at matching exact keywords, it falls short of capturing the semantic meaning of the claim and evidence sentences. Hence, a sentence transformer was then leveraged to create embeddings and compute the cosine similarity between the claim and evidences, as we learned from the implementation of FAISS. This will be further discussed in Section 3.

A few sentence transformer models were also experimented including 'all-MiniLM-L6-v2', 'all-distilroberta-v1' and 'all-mpnet-base-v2'. Each model's results are discussed in the results section. The list of evidences were then ranked according to the cosine similarities and the top 10 candidates with the highest score was selected to be used for the classification aspect. This novel approach addresses the challenge of capturing both the lexical and semantic implications of the tokens in the claim and evidence sentences.

2.5 roBERTa

We trained and evaluated two classifiers from the roBERTa family:

- roBERTa-base: 12-layer (125 M parameters)
- distilroBERTa-base: A 6-layer (82 M parameters) model distilled from the RoBERTa model roberta-base checkpoint.

2.6 Evidence Fusion Verifier

To overcome the limitations of independent claim–evidence classification, we designed the **Evidence Fusion Verifier (EFV)**, a single-pass model that jointly reasons over multiple retrieved snippets. EFV builds directly on a shared DistilRoBERTa backbone but adds a lightweight fusion module to capture cross-snippet interactions.

Shared Encoder We begin by encoding both the claim and each of its k retrieved evidence passages using a single pretrained Transformer encoder, `distilroberta-base`. Given a batch of B claims, tokenised as `claim_input_ids` and `claim_attention_mask` of shape (B, L), the encoder produces hidden states from which we extract the [CLS] embedding for each claim:

$$c_i = \text{Encoder}\big(\texttt{claim\_input\_ids}_i,\\ \texttt{claim\_attention\_masks}_i\big)[\text{CLS}]$$

yielding a tensor $\mathbf{C} \in \mathbb{R}^{B\times d}$ with $d = 768$.

Evidence Encoding Each of the k evidence snippets for the i-th claim is similarly tokenised into `evidence_input_ids` of shape $(B,\ k,\ L)$ and flattened to $(B \cdot k,\ L)$ for efficient batch encoding. We again extract the [CLS] embeddings and reshape back to

$$\mathbf{E} \ \in \ \mathbb{R}^{B\times k\times d},$$

so that $\mathbf{e}_{i,j}$ is the embedding of the j-th evidence for claim i.

Transformer Fusion To allow each evidence to attend both to the claim and to other evidences, we stack the $k+1$ embeddings into a sequence

$$\big[\mathbf{c}_i,\ \mathbf{e}_{i,1},\ \mathbf{e}_{i,2},\ \ldots,\ \mathbf{e}_{i,k}\big] \ \in \ \mathbb{R}^{(k+1)\times d}$$

and feed it through a small Transformer encoder with 2 layers, 8 attention heads, and a 4× feed-forward expansion. This fusion module, derived from ideas in UnifEE's graph-Transformer over evidence nodes (Hu et al., 2023), updates each node embedding based on self-attention across all $k+1$ positions.

Classification Head After fusion, we pool the updated claim position (the first token) to obtain a final representation

$$\tilde{\mathbf{c}}_i \ = \ \text{Fusion}\big([\mathbf{c}_i, \mathbf{E}_i]\big)\,[0] \ \in \ \mathbb{R}^d,$$

and apply a linear layer to predict one of the four labels (`SUPPORTS`, `REFUTES`, `NOT_ENOUGH_INFO`, `DISPUTED`),

$$\text{logits}_i \ = \ \mathbf{W}\,\tilde{\mathbf{c}}_i + \mathbf{b} \quad \in \mathbb{R}^4$$

Our design draws on the **Dual-Channel Unified Format (DCUF)** model from NAACL2022, which processes heterogeneous evidence streams (e.g. text and table modalities) in parallel encoder channels before fusing their outputs via concatenation and a classification head. (Hu et al., 2022) While DCUF targets multimodal data, its core insight, *learned fusion of multiple representation channels* motivated our choice to fuse multiple textual evidence embeddings via a transformer rather than treating each independently. We simplified DCUF's dual-channel concept into a *single shared encoder* plus a *Transformer-based fusion module*, making the model lighter and fully Colab-compatible.

Additionally, the UnifEE graph-Transformer approach (Hu et al., 2023) showed the effectiveness of self-attention over evidence nodes; EFV implements this in a more tractable form by stacking embeddings rather than constructing explicit graphs. Together, these inspirations guided our EFV design toward a balance of performance, simplicity, and resource efficiency.

3 Experiments

3.1 Evaluation method

3.1.1 Dataset

We evaluated our system on a climate-science specific fact-verification corpus provided by the COMP90042: Natural Language Processing teaching team that contains pre-split training and development sets.

3.1.2 Evaluation Metrics

Defined by COMP90042 teaching team:

- 1. Evidence Retrieval F-score (F)
- 2. Claim Classification Accuracy (A)
- 3. Harmonic Mean of F and A

3.2 Experimental details

We evaluated four models on our claim-verification task: two RoBERTa variants *RoBERTa-Base* and *DistilRoBERTa-Base*) and our custom *Evidence Fusion Verifier* (EFV). All experiments were run on a free-tier Colab T4 GPU (12 GB RAM) using PyTorch and the HuggingFace Transformers library.

For **roBERTa-base**, we used a small per-device batch size of 8 for both training and evaluation, training for just two epochs. We evaluated and logged metrics at the end of each epoch and disabled checkpoint saving to reduce overhead. To compensate for the small batch size, we applied `gradient_accumulation_steps=2`, effectively simulating a batch size of 16. Learning proceeded with a rate of 2×10^{-5} and a weight decay of 0.01, balancing convergence speed against regularisation.

For **distilroBERTa-base**, we increased the per-device batch size to 16 and extended training to ten epochs, reflecting the model's lighter memory footprint. We set weight decay to 0.05 and configured the Trainer to evaluate, save, and log at each epoch. Crucially, we enabled `load_best_model_at_end` using macro-F_1 as the selection metric, and attached an `EarlyStoppingCallback` with patience of two epochs.

For the **Evidence Fusion Verifier (EFV)**, we built on the **distilroBERTa-base** encoder and added a two-layer Transformer fusion module (hidden size 768, eight attention heads) to jointly process one claim embedding along with $k = 5$ evidence embeddings in a single forward pass. We fine-tuned this model with a learning rate of 2×10^{-5}, weight decay of 0.05, and sequences truncated to length 256. Training used a per-device batch size of 16 for up to ten epochs, with an early-stopping patience of two epochs on the development set macro-F_1. In practice, the fusion verifier reached its best macro-F_1 at epoch 5 and halted thereafter, completing in approximately ten minutes on a Colab T4 GPU.

All learning rates and weight-decay values were selected via small grid searches: higher learning rates caused unstable loss curves, while lower rates slowed convergence unduly. Despite adding only $\sim$1 M parameters, the EFV achieved a 4-point improvement in macro-F1 over the best single-pair DistilRoBERTa model, demonstrating that multi-evidence fusion is both efficient and effective under our resource constraints.

3.3 roBERTa Model Selection Using Gold Evidence

To establish a baseline, we fine-tuned two RoBERTa variants on the *gold* evidence (i.e. the ground-truth passages, not those retrieved by our hybrid system). Both models used identical settings (learning rate 2×10^{-5}, weight decay 0.05, batch size 16, max sequence length 256, early stopping on macro-F_1 with patience=2). On the development set with gold evidence, RoBERTa-Base reached a macro-F_1 of 0.54 (in 3 epochs) and DistilRoBERTa-Base reached 0.49 (in 6 epochs). Given the small 5-point F_1 gap but significantly lower memory and inference costs, we chose DistilRoBERTa-Base for all subsequent experiments.

4 Result

4.1 Final End-to-End Evaluation

Our final end-to-end pipeline pairs the Hybrid Retriever with the Evidence Fusion Verifier to achieve robust claim classification despite fuzzy evidence retrieval. To assess our complete pipeline under the official COMP90042 evaluation script (`eval.py`), we measured three key metrics on the dev set:

Metric	Value
Evidence Retrieval F_1 (F)	0.05500
Claim Classification Accuracy (A)	0.33116
Harmonic Mean (H) of F and A	0.09434

Table 1: Key evaluation metrics.

Despite the rather low retrieval F_1, a reflection of the difficulty in retrieving all gold evidence sentences, our Evidence Fusion Verifier proved robust to incomplete or noisy input. By attending over whichever passages were available (even when many gold snippets were missed), EFV still extracted partial signals that enabled reasonable classification performance. In other words, while perfect retrieval remains an aspirational goal, the fusion-based verifier mitigates this bottleneck by aggregating weak or partial evidence into a coherent judgment.

4.2 Comparison between roBERTa and EFV

Both the Evidence Fusion Verifier (EFV) and a standard DistilRoBERTa classifier use the same distilroberta-base backbone, [CLS]–claim–[SEP]–evidence–[SEP] encoding, Transformers fine-tuning interface, and a linear head to predict SUPPORTS, REFUTES, NOT_ENOUGH_INFO, or DISPUTED. A vanilla DistilRoBERTa independently encodes each of the k retrieved passages, incurring k forward passes and heuristic aggregation, whereas EFV ingests the claim and all k snippets in one pass, stacks their [CLS] embeddings into a sequence, and applies a lightweight fusion Transformer to capture inter-snippet interactions.

We thought that the architectural change to the EFV would bring several practical advantages. First, cross-evidence reasoning becomes possible. The model can attend from one snippet to another, identifying when two pieces of information must be combined to validate a multi-step claim. Second, because aggregation is internalised, the overhead of k separate inference calls are avoided. This often leads to faster end-to-end latency when k is small. Third, real-world retrieval is noisy: the EFV can learn to down-weight irrelevant passages through its attention mechanism, whereas a vanilla classifier has no built-in way to ignore "distractor" evidence other than by chance.

4.3 Analysis of Classifier Discrepancies

When both DistilRoBERTa and EFV are applied to the *same* set of hybrid-retrieved evidence snippets, their per-class precision, recall, and F_1 metrics differ.

Class	DR+Hybrid F_1	EFV+Hybrid F_1
SUPPORTS	0.66	0.62
REFUTES	0.33	0.24
NOT_ENOUGH_INFO	0.47	0.45
DISPUTED	0.33	0.44

Table 2: Per-class F_1 comparison between DistilRoBERTa (DR) and EFV on hybrid retrieval.

Architectural Precision–Recall Trade-offs DistilRoBERTa processes each snippet in isolation and aggregates externally (e.g. majority vote), yielding high recall on easily supported claims but limited ability to detect cross-snippet contradictions. EFV employs a fusion Transformer to attend over all k snippets jointly, becoming more conservative (fewer false positives) and more precise at classes requiring multi-evidence reasoning.

However, our Hybrid retriever may return low relevance passages. DistilRoBERTa can be misled by out-of-context snippets, whereas EFV's self-attention fusion learns to down-weight noisy evidence, improving precision especially for `REFUTES` and `DISPUTED`.

4.4 EFV Setup and Improvements

In our current EFV setup we fix the number of evidence snippets per claim to $k = 5$ for both computational efficiency and to satisfy the 12 GB Colab memory limit. However, in practice this artificial cap may omit partially relevant passages that, although individually weak, collectively carry important signals. By increasing or removing the k constraint, the fusion Transformer could attend over a richer set of contexts, allowing it to:

- *Capture corroborating details* across more than five candidate passages, thereby boosting recall on classes like `SUPPORTS` and `NOT_ENOUGH_INFO`.
- *Better model conflicting evidence* for `DISPUTED` claims, since contradictions sometimes span more than five snippets.
- *Mitigate retrieval noise* by diluting the impact of any single irrelevant snippet, attending to dozens of weakly relevant passages tends to average out noise and emphasize consistent patterns.

In preliminary experiments we increased the evidence budget to $k = 10$ in order to expose the fusion module to a wider context. While this configuration yielded a modest macro-F_1 gain, particularly on `DISPUTED` claims, it also caused our Colab T4 GPU to run out of memory and crash during both training and inference. The jump from 5 to 10 passages per claim roughly doubled the sequence-batch footprint (from $(B \times 6) \times L$ to $(B \times 11) \times L$ token embeddings), exceeding the 12GB memory limit even with gradient accumulation and mixed-precision enabled. Thus, although a larger k can unlock additional cross-snippet signals, the current resource constraints force a trade-off between performance and stability, and we settled on $k = 5$ as the maximum that reliably fits within a free-tier GPU environment.

5 Conclusion

Our experiments on the COMP90042 predefined climate-science corpus demonstrated that lightweight models can achieve a relatively sound performance when the domain is constrained, like this educational case. However, our final pipeline has several constraints. Its end to end performance is eventually bounded by the quality of the hybrid retriever. Low quality passages cannot be fully remedied by the verifier. Moreover, the fixed $k = 5$ retrieved evidences to satisfy GPU memory limits have excluded useful evidence especially when more context was needed.

Looking ahead, several approaches could potentially strengthen our system: exploring graph-based or sequential multi-hop retrieval, integrating a lightweight cross-encoder to re-rank top candidates, or specific to this dataset, applying pretraining on climate science literature. Incorporating explainable rationale generation would also help understand the reasoning behind each verdict. Together, these enhancements can take us closer to a scalable, robust fact-checking system capable of handling both specialized domains and broader, real-world scenarios.

6 Team contributions

Justin:

- 1. Provided a high-level development guideline for the team
- 2. Implemented and evaluated both classification models
- 3. Discussed the different classification models in the report

Luis:

- 1. Preprocessed the data to be ready for analysis
- 2. Implemented and evaluated the hybrid retrieval model
- 3. Completed a first draft of the report

Toby:

- 1. Implemented and evaluated the baseline BM25 retrieval model and FAISS
- 2. Discussed the different retrieval models in the report
- 3. Edited the format and content of the final version of the report

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